

0406320 Solar PV Systems –Group Project (SAM)

INSTRUCTORS: DI ZHANG

Each group (2-4 students) must submit a PDF report document (subjected to SafeAssign plagiarism check) and the final SAM project file (.sam).

Download and install System Advisor Model (SAM) (<https://sam.nrl.gov/download.html>).

The main reference (only the residential part until 33min): <https://www.youtube.com/watch?v=ba5NcqlANZQ>

Design information:

- In “Location and resources”, use SAM’s built-in menu to download the weather data for Sharjah.
- Design a roof PV system for the **UOS male sports complex** (location: 25.284503799178378, 55.478224746237565), using the two tilted sides of the roof.
- Assume that the tilt of each side is 5°; PV modules will be installed flushed to the roof (no extra tilt).
- Use [Google Earth](#) to measure the roof azimuths and size.
- Follow the steps of the video reference to perform initial calculations; choose any appropriate models of PV modules and inverters; complete the system sizing by deciding the PV array and inverter configurations (you should implement 2 subarrays for the 2 roof sides, similar to the video reference).
- Review and discuss the design steps and result analysis in the report.



Report requirements:

- Cover page (topic, names and UIDs of all team members).
- Show Google Earth screenshot with the roof measurement; mark on this image the roof azimuth angles.
- Include the initial calculations, similar to the video reference (see below).
- For the final design, show a screenshot of the “System Design” page from SAM; also show the screenshot of the result summary page.
- Include the following system conclusions:
 - The total PV DC power (all modules): ____ kW.
 - Show the screenshot of the “Time series” from the SAM results (system generated power kW across the year); the peak system power is ____ kW.
 - In a year, which roof is more productive (N/S) and how much are their kWh productions? (hint: from the “Data tables” page in the SAM results).
 - For the whole system, which month is the most productive and how much is the production, and what about the least productive month? (hint: from the result summary page).
- Similar to the video reference, run the System Sizing macro, and show the screenshot of Table 2 and Figure 1 from the macro result; are there significant losses in your design?
- Show the screenshot of the “Losses” page from the SAM results; discuss the major losses and suggest ways to improve performance.

Initial calculations to be included in the report:

Design the system: System size

Use roof area to determine maximum system capacity:
Array Capacity (W) = Roof Area (m²) * Module Efficiency * 1,000 W/m²
 $35 \text{ m}^2 * 0.15 * 1,000 \text{ W/m}^2 = 5,250 \text{ W}$

Assume about 1,000 W/m² of sunlight available on a clear sunny day at noon based on Standard Test Conditions (STC).

About 17 x 12 feet available on each surface:
 $17 \text{ ft} * 12 \text{ ft}$
 $= 204 \text{ sq ft} * 2$
 $= 408 \text{ sq ft}$
 $= 35 \text{ m}^2$



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Design the system: Inverter

- ✓ System Size
~5 DC kW, 1.2 DC/AC ratio
- Inverter
 $5,000 \text{ DC W} / 1.2 = 4166.7 \text{ AC W} \approx 4200 \text{ AC W}$
- Module
- String Size
- Number of Strings
- Array Orientation

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Design the system: Module

- ✓ System Size
5 kW, 1.2 DC/AC ratio
- ✓ Inverter
4200 AC W Power-One PVI-4.2
- Module
Initial guess: 250 W modules, 5,000 W array / 250 W module = 20 modules
- String Size
- Number of Strings
- Array Orientation

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Design the system: String size

- ✓ System Size
5 kW, 1.2 DC/AC ratio
- ✓ Inverter
4200 AC W Power-One PVI-4.2
- ✓ Module
250 DC W Trina Solar TSM-250
- String Size
Inverter Maximum Voltage / Module Voc
Maximum modules per string: $480 \text{ VDC} / 37.8 \text{ VDC} = 12.7 \approx 12$
Minimum modules per string: $100 \text{ VDC} / 37.8 \text{ VDC} = 2.65 \approx 3$
- Number of Strings
- Array Orientation

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System design

- ✓ System Size
5 kW, 1.2 DC/AC ratio
- ✓ Inverter
4200 AC W Power-One PVI-4.2
- ✓ Module
250 DC W Trina Solar TSM-250
- ✓ String Size
10 modules
- ✓ Number of Strings
2 strings, 2 inverter MPPT inputs
- ✓ Array orientation
Two subarrays, 10 degree tilt, E-W facing

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